

PAPER_1347 — High-Tc Superconductivity Origin

Framework: UQFF — Star-Magic v5.27+ | **Tier:** F Condensed Matter (CLOSED) **Calculator:**
calculate_paradox({"paradox": "htsc"})

Master Expression

$$T_c = \hbar \cdot \omega_{SCm} / k_B \times K_{MEX} = 125 \text{ K}$$

Match: match cuprate YBCO/BSCCO range

Interpretation

SCm phonon resonance $\times$ K_{MEX} Mexican-hat couples to Cooper-pair buoyancy. No need for exotic boson mediators.

UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{CH} = 9$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$, $F_{TRZ} = 0.1$
- $\omega_{SCm} = 1.25 \text{ THz}$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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